

 dummy DataFrame as df

```
-- 1.1
```

```
with alert_counts as (
  select
    hcp_id,
    count(alert_id) as total_alerts,
    row_number() over (order by count(alert_id) desc) as rank
  from
    Alerts.csv
  group by
    hcp_id
)
select
  rank,
  hcp_id,
  total_alerts
from
  alert_counts
where
  rank <= 10;
```

index	...	↑↓	rank	...	↑↓	HCP_ID	...	↑↓	total_alerts
		0			1	HCP041			
		1			2	HCP011			
		2			3	HCP019			
		3			4	HCP020			
		4			5	HCP032			
		5			6	HCP021			
		6			7	HCP014			
		7			8	HCP036			
		8			9	HCP022			
		9			10	HCP007			

Rows: 10

 Expand

 dummy DataFrame as df1

-- 1.2

```
with filtered_sales as (
  select
    s.hcp_id,
    sum(s.prescription_volume) as total_prescription_volume,
    row_number() over (order by sum(s.prescription_volume) desc) as rank
  from
    Sales.csv s
  inner join (
    select distinct hcp_id
    from Alerts.csv
  ) a on s.hcp_id = a.hcp_id
  group by
    s.hcp_id
)
select
  rank,
  hcp_id,
  total_prescription_volume
from
  filtered_sales
where
  rank <= 10;
```

index	...	↑↓	rank	...	↑↓	HCP_ID	...	↑↓	total_prescription_volume
		0			1	HCP185			
		1			2	HCP062			
		2			3	HCP019			
		3			4	HCP163			
		4			5	HCP211			
		5			6	HCP126			
		6			7	HCP154			
		7			8	HCP215			
		8			9	HCP088			
		9			10	HCP004			

Rows: 10

 Expand

How likely are you to recommend DataLab to a friend or co-worker?

×

Not at all likely 0 1 2 3 4 5 6 7 8 9 10 Extremely likely

powered by InMoment

 dummy DataFrame as df2

```
-- 2
```

```
with positivealerts as (
  select
    hcp_id,
    count(alert_id) as total_positive_alerts,
    min(alert_date) as first_positive_alert_date
  from Alerts.csv
  where lower(lab_result) = 'positive'
  group by hcp_id
),
prescriptionswithin30days as (
  select distinct
    p.hcp_id
  from positivealerts p
  join Sales.csv s
    on p.hcp_id = s.hcp_id
  where s.prescription_date >= p.first_positive_alert_date
    and s.prescription_date <= p.first_positive_alert_date + interval '30 days'
)
select
  row_number() over (order by pa.total_positive_alerts desc) as rank,
  pa.hcp_id,
  pa.total_positive_alerts as "positive alerts"
from positivealerts pa
left join prescriptionswithin30days rx
  on pa.hcp_id = rx.hcp_id
where rx.hcp_id is null
order by pa.total_positive_alerts desc
limit 10;
```

index	...	↑↓	rank	...	↑↓	HCP_ID	...	↑↓	positive alerts
		0			1	HCP041			
		1			2	HCP011			
		2			3	HCP021			
		3			4	HCP042			
		4			5	HCP023			
		5			6	HCP007			
		6			7	HCP026			
		7			8	HCP009			
		8			9	HCP001			
		9			10	HCP002			

Rows: 10

[Expand](#)

How likely are you to recommend DataLab to a friend or co-worker?

Not at all likely 0 1 2 3 4 5 6 7 8 9 10 Extremely likely

powered by InMoment

dummy DataFrame as df3

-- 3.1

```
with top10alerthcps as (
  select
    hcp_id
  from Alerts.csv
  group by hcp_id
  order by count(alert_id) desc
  limit 10
),
hcp_mix as (
  select
    t.hcp_id,
    coalesce(sum(s.prescription_volume), 0) as total_prescription_volume,
    coalesce(sum(
      case
        when lower(s.drug_id) = 'drg001'
        then s.prescription_volume
        else 0
      end), 0) as our_drug_volume,
    coalesce(sum(
      case
        when lower(s.drug_id) <> 'drg001'
        then s.prescription_volume
        else 0
      end), 0) as competitor_volume
  from top10alerthcps t
  left join Sales.csv s
    on t.hcp_id = s.hcp_id
  group by t.hcp_id
)
select
  hcp_id,
  total_prescription_volume,
  case
    when total_prescription_volume = 0
    then '0.0%'
    else round(our_drug_volume * 100.0 / total_prescription_volume, 1)::text || '%'
  end as "% our drug share",
  case
    when total_prescription_volume = 0
    then '0.0%'
    else round(competitor_volume * 100.0 / total_prescription_volume, 1)::text || '%'
  end as "% competitor share"
from hcp_mix
order by
  case
    when total_prescription_volume = 0 then 0
    else competitor_volume * 100.0 / total_prescription_volume
  end desc;
```

...	↑↓	...	↑↓	total_prescription_volume	...	↑↓	% our drug ...	...	↑↓	% competitor s...	...	↑↓
0		HCP020		570			25.4%			74.6%		
1		HCP019		2277			59.1%			40.9%		
2		HCP032		2075			59.1%			40.9%		
3		HCP022		1363			62.0%			38.0%		
4		HCP036		1895			64.7%			35.3%		
5		HCP014		944			71.1%			28.9%		
6		HCP007		0			0.0%			0.0%		
7		HCP041		0			0.0%			0.0%		
8		HCP011		0			0.0%			0.0%		
9		HCP021		0			0.0%			0.0%		

Rows: 10

Expand

How likely are you to recommend DataLab to a friend or co-worker?

Not at all likely 0 1 2 3 4 5 6 7 8 9 10 Extremely likely

powered by InMoment

dummy DataFrame as d

```
-- 3.2
```

```
with top10alerthcps as (  
  select  
    hcp_id  
  from Alerts.csv  
  group by hcp_id  
  order by count(alert_id) desc, hcp_id  
  limit 10  
)  
select  
  s.drug_name as "competitor drug name",  
  s.drug_id,  
  sum(s.prescription_volume) as total_prescription_volume  
from Sales.csv s  
join top10alerthcps t  
  on s.hcp_id = t.hcp_id  
where lower(s.drug_id) <> 'drg001'  
group by  
  s.drug_name,  
  s.drug_id  
order by  
  sum(s.prescription_volume) desc,  
  s.drug_id  
limit 1;
```

...	↑↓	competitor drug n...	...	↑↓	...	↑↓	total_prescription_volume	...	↑↓
0		Rivalex			DRG002		1470		

Rows: 1

[Expand](#)

How likely are you to recommend DataLab to a friend or co-worker?

Not at all likely 0 1 2 3 4 5 6 7 8 9 10 Extremely likely

powered by InMoment

dummy DataFrame as d

-- 4

```
with accountalerts as (
  select
    af.account_id,
    count(al.alert_id) as total_alerts
  from Alerts.csv al
  inner join Affiliation.csv af
    on al.hcp_id = af.hcp_id
  group by af.account_id
),
top10accounts as (
  select
    account_id,
    total_alerts
  from accountalerts
  order by total_alerts desc, account_id
  limit 10
),
accountprescriptions as (
  select
    af.account_id,
    count(s.prescription_id) as our_drug_rx_count
  from Sales.csv s
  inner join Affiliation.csv af
    on s.hcp_id = af.hcp_id
  where lower(s.drug_id) = 'drg001'
  group by af.account_id
)
select
  row_number() over (
    order by coalesce(rx.our_drug_rx_count, 0) * 1.0 / aa.total_alerts asc
  ) as rank,
  aa.account_id,
  aa.total_alerts as "total alerts",
  coalesce(rx.our_drug_rx_count, 0) as "our drug prescription count",
  round(
    coalesce(rx.our_drug_rx_count, 0) * 100.0 / aa.total_alerts,
    1
  )::text || '%' as "conversion ratio"
from top10accounts aa
left join accountprescriptions rx
  on aa.account_id = rx.account_id
order by rank;
```

...	↑↓	...	↑↓	A...	...	↑↓	total ...	...	↑↓	our drug prescription count	...	↑↓	conversion r...	...	↑↓
0		1		ACC001			3296			385			11.7%		
1		2		ACC003			1195			159			13.3%		
2		3		ACC002			2350			373			15.9%		
3		4		ACC004			530			164			30.9%		
4		5		ACC005			585			275			47.0%		
5		6		ACC006			493			298			60.4%		
6		7		ACC012			85			57			67.1%		
7		8		ACC009			100			77			77.0%		
8		9		ACC007			114			209			183.3%		
9		10		ACC010			92			295			320.7%		

Rows: 10

Expand

How likely are you to recommend DataLab to a friend or co-worker?

Not at all likely 0 1 2 3 4 5 6 7 8 9 10 Extremely likely

powered by InMoment

 dummy DataFrame as

```
-- 5

with accountbase as (
  select
    account_id,
    count(distinct hcp_id) as total_affiliated_doctors
  from Affiliation.csv
  group by account_id
),
activeprescribers as (
  select
    a.account_id,
    count(distinct s.hcp_id) as active_doctors,
    sum(s.prescription_volume) as total_rx_volume
  from Sales.csv s
  inner join Affiliation.csv a
    on s.hcp_id = a.hcp_id
  where lower(s.drug_id) = 'drg001'
  group by a.account_id
)
select
  row_number() over (
    order by (ap.total_rx_volume * 1.0 / ap.active_doctors) asc
  ) as rank,
  ab.account_id,
  ab.total_affiliated_doctors as "# affiliated doctors",
  ap.active_doctors as "# active doctors",
  ap.total_rx_volume as "our drug prescription volume",
  round(ap.total_rx_volume * 1.0 / ap.active_doctors, 1)
    as "prescription volume per active doctor"
from accountbase ab
join activeprescribers ap
  on ab.account_id = ap.account_id
where ap.active_doctors > 0
order by "prescription volume per active doctor" asc
limit 10;
```

...	↑↓	...	↑↓	A...	...	↑↓	# affiliated doctors	...	↑↓	# active do...	...	↑↓	our drug prescription volume	...	↑↓	prescription volume p
0		1		ACC017			3			1			59			
1		2		ACC015			8			7			1347			
2		3		ACC012			10			7			1804			
3		4		ACC009			11			10			2699			
4		5		ACC003			24			15			4947			
5		6		ACC018			4			3			1095			
6		7		ACC004			17			13			5402			
7		8		ACC008			7			6			3049			
8		9		ACC014			6			6			3126			
9		10		ACC023			1			1			573			

Rows: 10

[Expand](#)

How likely are you to recommend DataLab to a friend or co-worker?

Not at all likely

0

1

2

3

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10

Extremely likely

powered by InMoment

 dummy DataFrame as

```
-- 6.1

with firstalerts as (
  select
    hcp_id,
    min(alert_date) as first_alert_date
  from Alerts.csv
  where lower(lab_result) = 'positive'
  group by hcp_id
),
saleswindows as (
  select
    fa.hcp_id,
    coalesce(sum(
      case
        when s.prescription_date >= fa.first_alert_date - interval '90 days'
        and s.prescription_date < fa.first_alert_date
        then s.prescription_volume
        else 0
      end
    ), 0) as pre_volume,
    coalesce(sum(
      case
        when s.prescription_date >= fa.first_alert_date
        and s.prescription_date <= fa.first_alert_date + interval '90 days'
        then s.prescription_volume
        else 0
      end
    ), 0) as post_volume
  from firstalerts fa
  left join Sales.csv s
    on fa.hcp_id = s.hcp_id
    and lower(s.drug_id) = 'drg001'
  group by fa.hcp_id
),
liftcalculation as (
  select
    hcp_id,
    pre_volume,
    post_volume,
    case
      when pre_volume > 0
      then ((post_volume * 1.0 - pre_volume) / pre_volume) * 100
      else null
    end as lift_pct
  from saleswindows
)
select
  row_number() over (order by lift_pct desc) as rank,
  hcp_id,
  pre_volume as "total prescription volume before alert",
  post_volume as "total prescription volume after alert",
  '+' || round(lift_pct, 1)::text || '%' as "lift %"
from liftcalculation
where lift_pct is not null
order by lift_pct desc
limit 10;
```

...	↑↓	...	↑↓	...	↑↓	total prescription volume before alert	...	↑↓	total prescription volume after alert	...	↑↓	...	↑↓
0		1		HCP040				8			143		+1687.5%
1		2		HCP050				10			107		+970.0%
2		3		HCP006				32			249		+678.1%
3		4		HCP108				31			238		+667.7%
4		5		HCP016				50			279		+458.0%
5		6		HCP159				42			232		+452.4%
6		7		HCP157				72			387		+437.5%

How likely are you to recommend DataLab to a friend or co-worker?

Not at all likely

0

1

2

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10

Extremely likely

```

dummy DataFrame as

-- 6.2

with firstalerts as (
  select
    hcp_id,
    min(alert_date) as first_alert_date
  from Alerts.csv
  where lower(lab_result) = 'positive'
  group by hcp_id
),
saleswindows as (
  select
    fa.hcp_id,
    coalesce(sum(
      case
        when s.prescription_date >= fa.first_alert_date - interval '90 days'
        and s.prescription_date < fa.first_alert_date
        then s.prescription_volume
        else 0
      end
    ), 0) as pre_volume,
    coalesce(sum(
      case
        when s.prescription_date >= fa.first_alert_date
        and s.prescription_date <= fa.first_alert_date + interval '90 days'
        then s.prescription_volume
        else 0
      end
    ), 0) as post_volume
  from firstalerts fa
  left join Sales.csv s
    on fa.hcp_id = s.hcp_id
    and lower(s.drug_id) = 'drg001'
  group by fa.hcp_id
),
hcp_segments as (
  select
    hcp_id,
    pre_volume,
    post_volume,
    (post_volume - pre_volume) as absolute_growth,
    case
      when pre_volume > 0
      then ((post_volume * 1.0 - pre_volume) / pre_volume) * 100
      else null
    end as lift_pct,
    case
      when pre_volume = 0
      and post_volume > 0
      then 'new starters'
      when pre_volume > 0
      and ((post_volume * 1.0 - pre_volume) / pre_volume) * 100 >= 20
      then 'growers'
      else 'non-responders'
    end as segment
  from saleswindows
)
select
  coalesce(segment, 'total') as segment,
  count(hcp_id) as hcps,
  case
    when avg(lift_pct) is null
    then 'n/a'
    else '+' || round(avg(lift_pct), 1)::text || '%'
  end as "avg lift",
  sum(absolute_growth) as "total absolute growth (commercial impact)"
from hcp_segments
group by grouping sets

```

How likely are you to recommend DataLab to a friend or co-worker?

Not at all likely 0 1 2 3 4 5 6 7 8 9 10 Extremely likely

```
else 0
end,
segment;
```

...	↑↓	segment	...	↑↓	...	↑↓	...	↑↓	total absolute growth (commercial i...	...	↑↓	
0		growers			25		+322.8%				3319	
1		new starters			64		n/a				9516	
2		non-responders			125		+/-68.2%				-4623	
3		total			214		+69.5%				8212	

Rows: 4

Expand

How likely are you to recommend DataLab to a friend or co-worker?

Not at all likely

0

1

2

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Extremely likely

powered by InMoment

 dummy DataFrame as

```
-- 7.1
with params as (
  select
    40 as total_reps,
    20 as field_days_per_month,
    3 as months_in_quarter,
    8 as calls_per_day,
    0.45 as new_starter_weight,
    0.15 as grower_weight,
    0.40 as non_responder_weight
),
capacity as (
  select *,
    total_reps * field_days_per_month * months_in_quarter * calls_per_day
    as total_capacity
  from params
),
segmentdata as (
  select 'new starters' as segment, 64 as hcp_count
  union all
  select 'growers', 25
  union all
  select 'non-responders', 125
)
select
  segment,
  hcp_count as "hcps",
  case
    when segment = 'new starters'
      then round(new_starter_weight * 100, 0)::text || '%'
    when segment = 'growers'
      then round(grower_weight * 100, 0)::text || '%'
    when segment = 'non-responders'
      then round(non_responder_weight * 100, 0)::text || '%'
  end as "suggested % of rep calls",
  case
    when segment = 'new starters'
      then cast(total_capacity * new_starter_weight as integer)
    when segment = 'growers'
      then cast(total_capacity * grower_weight as integer)
    when segment = 'non-responders'
      then cast(total_capacity * non_responder_weight as integer)
  end as "estimated quarterly calls",
  case
    when segment = 'new starters'
      then cast(round(total_reps * new_starter_weight, 0) as integer)
    when segment = 'growers'
      then cast(round(total_reps * grower_weight, 0) as integer)
    when segment = 'non-responders'
      then cast(round(total_reps * non_responder_weight, 0) as integer)
  end as "estimated rep allocation"
from segmentdata
cross join capacity
union all
select
  'total',
  sum(hcp_count),
  '100%',
  max(total_capacity),
  max(total_reps)
from segmentdata
cross join capacity;
```

...	↑↓	segment	...	↑↓	...	↑↓	suggested % of rep calls	...	↑↓	estimated quarterly calls	...	↑↓	estimated rep allocation	...	↑↓
0		new starters			64		45%			8640			18		
1		non-responders			125		40%			7680			16		
2		growers			25		15%			2880			6		

How likely are you to recommend DataLab to a friend or co-worker?

Not at all likely 0 1 2 3 4 5 6 7 8 9 10 Extremely likely

powered by InMoment

 dummy DataFrame as

-- 7.2

```
with params as (  
  select  
    40 as total_reps,  
    20 as field_days_per_month,  
    3 as months_in_quarter,  
    8 as calls_per_day  
)  
select  
  total_reps * field_days_per_month * months_in_quarter * calls_per_day  
  as "total rep call capacity per quarter"  
from params;
```

...	↑↓	total rep call capacity per quarter	...	↑↓	
		0			19200

Rows: 1 ↗ Expand

How likely are you to recommend DataLab to a friend or co-worker?

Not at all likely

0

1

2

3

4

5

6

7

8

9

10

Extremely likely